

João Serra

CURRICULUM VITAE | SEPTEMBER 2026

João Guilherme Valente Mendes Serra

Year of birth: 1993

Affiliation: Land-CRAFT, Aarhus University, Denmark

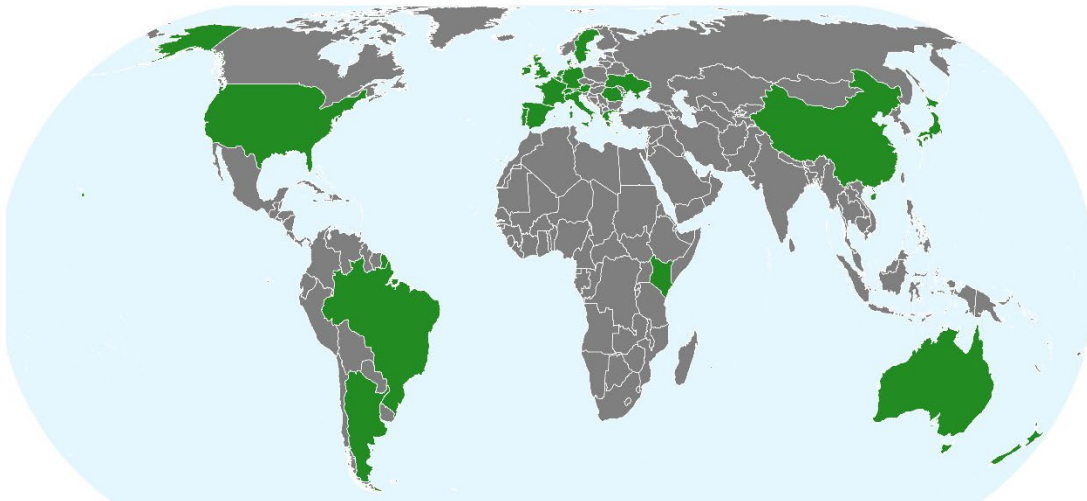
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Research areas

Agricultural nitrogen cycling and use efficiency; irrigation and water quality; greenhouse gas emissions; spatial analysis; process-based modelling and interpretable machine learning.

My scientific collaborations (2025)



Academic appointments and education

2024–present | Postdoctoral researcher

Department of Agroecology and Land-CRAFT, Aarhus University, Denmark.

2020–2024 | PhD in Environmental Sciences

Instituto Superior de Agronomia, Portugal; research also conducted at the Department of Agroecology, Aarhus University, Denmark. Degree awarded in 2024.

2018–2020 | Research fellow

NitroPortugal H2020: Strengthening Portuguese Research and Innovation Capacities in the Field of Excess Reactive Nitrogen.

Scientific leadership and expert groups

2025–present | FAO Cropland Nutrient Balance team

Member. <https://www.fao.org/faostat/en/#data/ESB>

2024–present | EU Nitrogen Expert Panel

Member and Leader of the Crop Working Group. <https://www.eunep.com/>

2025–present | DairyPathS

Leader of WP 3.4: modelling landscape scenarios for dairy production in living labs.

Peer review

2020–present. Reviewer for Nature Communications (IF 14.7), Communications Earth & Environment (IF 8.4), Ambio (IF 5.7), Sustainable production and consumption (IF 9.2), Earth's Future (IF 8.2), Agronomy for Sustainable Development (IF 8.2), Agricultural Water Management (IF 6.7), Science of the Total Environment (IF 9.8), Environmental Sciences Europe (IF 5.9). Environmental Science and Pollution Research (IF 5.8) and others.

Publications

Peer reviewed journals

2026

1. **Serra, J.**, Lassaletta, L., Abalos, D., Rahimi, J., Sanz-Cobena, A., Quemada, M., Kraus, D., Treat, C., Butterbach-Bahl, K., 2026. Irrigation as a coupled water-nitrogen flux: Correcting a blind spot in N₂O emissions. *One Earth* 9(9), 101808. doi:10.1016/j.oneear.2026.101808
2. Marinheiro, J., **Serra, J.**, Topp, C., Marques-dos-Santos, C., Fonseca, A., 2026. Resource-use archetypes in contrasting agrosilvopastoral systems: an emergy study of Portuguese Montado and Scottish Grasslands. *Agricultural Systems* 240, 104979. doi:10.1016/j.agry.2026.104979
3. **Serra, J.**, Lassaletta, L., Ros, G.H. et al., 2026. Global analysis of nitrogen inputs from irrigation water to cropland. *Nature Sustainability* doi:10.1038/s41893-026-01930-8
4. **Serra, J.**, Haas, E., Abalos, D., Butterbach-Bahl, K. et al, 2026. From patterns to decisions: detecting crop sequence and their drivers with explainable AI at field level in Denmark. *Agricultural Systems*. doi:10.1016/j.agry.2026.104982
5. Butterbach-Bahl, K., Philippot, L., **Serra, J.**, Silver, W. L., Ogle, S., Abalos, D., 2026. Three Decades of Soil N₂O Research, Insights and Gaps. *Global Change Biology* 32, no. 8, e71038. doi:10.1111/gcb.71038
6. Aderle, M. O., Haas, E., Liu, L., **Serra, J.**, Kraus, D., Butterbach-Bahl, K., Rahimi, J., 2026. mLDNDCv1.0: a machine learning-based surrogate of LandscapeDNDC for optimising cropping systems in Denmark. *Geoscientific Model Development* 19(13), 6335-6356. doi:10.5194/gmd-19-6335-2026
7. Bai, X., Cripps, T., **Serra, J.**, Butterbach-Bahl, K., Yao, Z., 2026. Spatial and temporal variability of CO₂, N₂O and CH₄ fluxes from an urban park in Denmark. *Biogeosciences* 23 (9), 3207-3223. doi:10.5194/bg-23-3207-2026
8. **Serra, J.**, Marinheiro, J., Marques-dos-Santos, C., Graversgaard, M., Quemada, M., Lassaletta, L., Ros, G., Giannini-Kurina, F., De Vries, W., Aguilera, E., Van Grinsven, H. J. M., Hayashi, K., Mogollon, J., Sanz-Cobena, A., Dalgaard, T., Butterbach-Bahl, K., 2026. Evaluation of global research on agricultural nitrogen pollution between 1990 and 2023: challenges for more efficacy and equity. *Environmental Science & Policy*, doi:10.1016/j.envsci.2026.104312
9. Marinheiro, J., Fonseca, A. M., Marques-dos-Santos, C., **Serra, J** (2026). Assessing Agricultural Systems using Emergy Analysis: A Review. *Agronomy* 15(9), 2110. doi:10.3390/agronomy15092110

2019 to 2025

10. Giannini-Kurina, F., **Serra, J.**, Christensen, B. T., Eriksen, J., Hutchings, N. J., Olesen, J. E., Jensen, J. L. (2025). Modelling and validating soil carbon dynamics at the long-term plot scale using the rCTOOL R package. *Environmental Modelling & Software* 183, 106229. doi:10.1016/j.envsoft.2024.106229

11. **Serra, J.**, Cameira, Marques-dos-Santos, C., Lassaletta, L., Aguilera, E., Sanz-Cobena, A., Garnier, J., de Vries, W., Dalgaard, T., Hutchings, N. J., Einarsson, R., Graversgaard, M., Medinets, S., Quemada, M. (2024). Assessing groundwater nitrate hotspots in Europe reveals severely inadequate Nitrate Vulnerable Zones. *Chemosphere* 355, 141830. doi:10.1016/j.chemosphere.2024.141830
12. **Serra, J.**, Cameira, M. R., Marques-dos-Santos, C., Lassaletta, L., Aguilera, E., Sanz-Cobena, A., Garnier, J., Billen, G., de Vries, W., Marinheiro, J., Dalgaard, T., Hutchings, N. J. (2023). Nitrogen Inputs by Irrigation is a Missing Link in the Agricultural Nitrogen Cycle and Related Policies in Europe. *Science of the Total Environment* 889, 164249. doi:10.1016/j.scitotenv.2023.164249
13. Garnier, J., Billen, G., Aguilera, E., Lassaletta, L., **Serra, J.**, Cameira, M. R., Marques-dos-Santos, C., Sanz-Cobena, A. (2023). How much can changes in the agro-food system reduce agricultural nitrogen losses to the environment? Example of a Temperate-Mediterranean gradient. *Journal of Environment Management* 337, 117732. doi:10.1016/j.jenvman.2023.117732
14. **Serra, J.**, Cameira, M. R., Marques-dos-Santos, C., Cruz, S., Hutchings, N. J. (2023). Is irrigation water an overlooked source of nitrogen in agriculture? *Agricultural Water Management* 278, 108147. doi:10.1016/j.agwat.2023.108147
15. Ragkos, A., Kentaro, H., **Serra, J.**, Shibata, H., Michalis, E., Eguchi, S., Oita, A., Marques-dos-Santos, C. (2021). Contrasting Considerations among Agricultural Stakeholders in Japan on Sustainable Nitrogen Management. *Sustainability* 13, 9:4866. doi:10.3390/su13094866
16. **Serra, J.**, Cameira, M. R., Marques-dos-Santos, C., & Hutchings, N. J. (2021). Development of a groundwater contamination index based on the agricultural hazard and aquifer vulnerability: Application to Portugal. *Science of the Total Environment* 772, 145032. doi:10.1016/j.scitotenv.2021.145032
17. **Serra, J.**, Marques-dos-Santos, C., Cameira, M. R., Cruz, S., & Hutchings, N. J. (2019). Challenges and solutions in identifying agricultural pollution hotspots using gross nitrogen balances. *Agriculture, Ecosystems & Environment*, 283, 106568. doi:10.1016/j.agee.2019.106568

Journal manuscripts under review

1. **Serra, J.**, Graversgaard, M., Quemada, M., Lassaletta, L., Ros, G., Giannini-Kurina, F., De Vries, W., Rahimi, J., Dalgaard, T., Butterbach-Bahl, K., (Under Review in *Nature Sustainability*). Cropland is trapped outside a safe and viable nitrogen use. <https://www.researchsquare.com/article/rs-8249505/v1>
2. Witt, N., **Serra, J.**, Wassénus, E., Graversgaard, M., Thorsoe, M. H. (Under Review in *One Earth*). Assessing corporate responsibility for nitrogen pollution: A tiered framework for ESG disclosures in agrifood supply chains. Available at SSRN: doi:10.2139/ssrn.6634000

3. Rahimi, J., **Serra, J.**, Ayedun, F. M., Tang, X., Liu, H., Rahimi, A., Barbacias, M. G., Haas, E., Petersen, S. O., Abalos, D., Butterbach-Bahl, K., (Under Review in Agriculture, Ecosystems & Environment). Nitrous oxide emissions from Danish agricultural systems: empirical insights and a machine learning re-interpretation of key drivers.
4. Frank, S., D., **Serra, J.**, Butterbach-Bahl, K. (Under Review in Environmental Modelling & Software). Spatial Optimization for Environmental Decision Support: A Systematic Review of Multi-Objective Frameworks for Agricultural Landscape Management.

Assessment chapters and guidance in production

1. **Serra, J.**, Medinets, S., Lassaletta, L., Zhang, X., Boincean, B., Aguilera, E. (In production). Missing inputs and outputs. In: Lassaletta, L., Sanz-Cobena, A. (eds). Guidance document on nitrogen use efficiency indicators for different purposes.
2. Garnier, J. and Sanz-Cobena, A. et al. (In production). Chapter 21: Assessment of nitrogen flows, impacts and solutions in West Europe. In: Sutton et al. (Under review) International Nitrogen Assessment.
3. Marques-dos-Santos, C., Runhaar, H., **Serra, J.**, et al, (In production). Chapter 25: Addressing the barriers to better nitrogen management. In: Sutton et al. (Under review) International Nitrogen Assessment.
4. Medinets, S., Moklyachuk et al., (In production). Chapter 22: Assessment of nitrogen flows, impacts and solutions in Eastern Europe. In: Sutton et al. (Under review) International Nitrogen Assessment.

Book chapters

1. Marinheiro, J., Fonseca, A. M., **Serra, J.** Marques-dos-Santos, C., 2023. Emergy Approach to the Sustainable Use of Ecosystems toward Better Land Management. In Agroecological Approaches for Sustainable Soil Management (eds M.N.V. Prasad and C. Kumar). doi:10.1002/9781119911999.ch9
2. Marques-dos-Santos, C., **Serra, J.**, Attard, G., Marchaim, U., Calvet, S., Amon, B., 2023. Available Technical Options for Manure Management in Environmentally Friendly and Circular Livestock Production. In: Bartzanas, T. (eds) Technology for Environmentally Friendly Livestock Production. Smart Animal Production. Springer, Cham. doi:10.1007/978-3-031-19730-7_7
3. Marques-dos-Santos, C., Marinheiro, J., **Serra, J.**, Cruz, S., Palmer, E., Hicks, K., Erisman, J. (2021). Nitrogen Footprints and the Role of Soil Enzymes. In: Moura, J. J. G., Moura, I., Maia, L. B. (eds) Enzymes for Solving Humankind's Problems. Springer, Cham. doi:10.1007/978-3-030-58315-6_5
4. Marques-dos-Santos, C., Bittman, S., Brito, L. M., Goss, M. J., Hunt, D., **Serra, J.**, Hutchings, N. (2020). Climate-resilient and smart agricultural management tools to cope with climate change-induced soil quality decline. Climate Change and Soil Interactions, 613–662. doi:10.1016/B978-0-12-818032-7.00022-9

Research projects

2026–present | GATE

Enabling the Green Transition through Federated, Open-Access Data Infrastructure. Gateway to agricultural and environmental data in Denmark. <https://gate.dti-skejby.dk/>

2026–present | AgNUE

Agricultural Nitrogen Use Efficiency Platform. Quantifying the impact of crop- and rotation-specific nitrogen use efficiency. <https://agnue.com/>

2025–present | DairyPathS

Agroecological Pathways for Sustainable, resilient and diversified Dairy farming systems in Europe assisted by multicriteria assessment. Leader of WP 3.4: modelling landscape scenarios for dairy production in living labs.

2025–present | iN-Net

International Nitrogen Network. Member of “Data” Working Group. Compiling datasets on nitrogen management in agriculture and applying agri-environmental indicators. <https://www.in-net.network/>

2025–present | RAINS

Resilient Agricultural Irrigation systems for water Scarcity in Europe. Modelling and spatial analysis of nitrogen and water flows in Mediterranean cropping systems. <https://rains-project.eu/>

2024–present | WHEP

Who has eaten the planet (ERC-2023-STG). Spatial preparation of key input data; water, soil and livestock modelling of nitrogen flows from pre-industrial periods to the present.

2024–present | PREMIS and SmartField

PREMIS: Primary activity data for emission inventories in farm accounts. Activities across PREMIS and SmartField: bottom-up farm-scale modelling of manure management systems and LandscapeDNDC experiments to assess drivers of nitrogen leaching in Denmark. Additional SmartField activities: machine learning analysis of agricultural N₂O emissions in Denmark and the role of irrigation in soil N₂O emissions. <https://smartfield.dk/>

2022–2025 | NOTS

NitroOrganicToSoils: improving nitrogen use efficiency and soil organic matter sequestration to reduce nitrogen fertilisation and soil inputs and improve soil quality. Responsibilities: reducing nitrogen losses to preserve water quality; qualification of farm indicators; supporting estimates of nitrogen fixation by legumes in Portugal; supporting a website to assist farmers’ management.

2020–2024 | MIXED

Multi-actor and transdisciplinary development of efficient and resilient mixed farming and agroforestry systems. Contribution to the Portuguese Montado mixed-farming case study.

2020–2023 | Aquifer-SUDOE

Innovative instruments for integrated groundwater management under increasing scarcity of hydrological resources. Good practices for water protection; field experiments for groundwater monitoring and protection; quantification of irrigation-water nitrogen inputs in Portugal, Spain and France.

2020–2023 | AgroGreen-SUDOE

Transnational cooperation to design more sustainable agricultural production strategies in the SUDOE Territory. Irrigation management using AquaCrop and modelling nitrate contamination in groundwater.

2019–2023 | INMS

Targeted Research for improving understanding of the global nitrogen cycle towards the establishment of an International Nitrogen Management System. Activity 1.1: nitrogen system indicators, including first authorship of a chapter and co-lead authorship of the NUE guidance document. Activity 1.6: barriers to better nitrogen management. Lead author of Chapter 25 and co-author of Chapter 21, focused on Northwestern Europe. <https://www.inms.international/>

2018–2020 | NitroPortugal H2020

Strengthening Portuguese Research and Innovation Capacities in the Field of Excess Reactive Nitrogen. Development of historical nitrogen flows in Portugal across air, soil and water; data management; deliverable reports.

Modelling and research software

Modelling expertise

Process-based models

LandscapeDNDC, AquaCrop-OSPy

Statistical and machine learning methods

Multivariate analysis, spatial clustering, machine (LightGBM, XGBoost, Random Forests) and deep (TabNet, neural networks) learning pipelines. Model interpretation via Explainable Artificial Intelligence (XAI) using SHAP values and partial dependence plots.

Emissions inventories

Empirical approaches for crop–livestock systems at national and subnational scales in Denmark and Portugal, including IPCC 2019 and EMEP/EEA methods for national greenhouse gas inventories.

Uncertainty and sensitivity analysis

Monte Carlo simulations with Latin hypercube sampling and variance-based sensitivity analysis using Sobol indices.

Research software

R | rCTOOL

Carbon turnover model for cropland soils, developed jointly with Dr. Franca Giannini-Kurina, Aarhus University. <https://github.com/francagiannini/rCTOOL>

R | emepManure

Inventory-based estimation of NH₃, N₂O and NO_x emissions from manure management systems. <https://github.com/jgserra18/emepManure>

R | livCH4ipcc

Inventory-based estimation of CH₄ emissions from enteric fermentation and manure management systems. <https://github.com/jgserra18/livCH4ipcc>

Python | pyLDNDC

Internal tool to configure and run LandscapeDNDC. <https://ldndc.imk-ifu.kit.edu/>

Python | pyFarmAC

Refactoring of Aarhus University's FarmAC model in Python. <https://github.com/jgserra18/pyFarmAC>

Python | pyglobwb

Development of the FAO global water balance model and formalisation as a Python package. <https://github.com/jgserra18/pyglobwb>

Research data

Published datasets alongside with publications (where possible) are available through Figshare (¹ https://figshare.com/authors/Jo_o_Serra/14259032), including scripts.

Research visit

June–July 2023 | Aarhus University

One-month visit to the Department of Agroecology. Parameterization of a grass–clover nitrogen conceptual model to estimate N₂O emissions, with Dr. Nicholas Hutchings and Prof. Jeroen Pullens. <https://github.com/jgserra18/grassclover>

Teaching and student supervision

University teaching

2023–2026 | Integrated Nutrient Management summer school

Department of Agroecology, Aarhus University, in collaboration with Hokkaido University, Japan. Lecturer on the impact of agronomic practices on groundwater nitrogen pollution; assistance with lectures and student group work. Academic contacts: Prof. Tommy Dalgaard and Dr. Morten Graversgaard.

2023/2024 | Environmental Monitoring

Instituto Superior de Agronomia, University of Lisbon. Theory classes on environmental monitoring and water quality, and European water-quality regulations: Water Framework Directive, Nitrates Directive and Groundwater Directive.

2023/2024 | Environment Resources and Society

Instituto Superior de Agronomia, University of Lisbon. Theory class on connections between agriculture and society.

2021/2022 and 2022/2023 | Environmental Pollution

Instituto Superior de Agronomia, University of Lisbon. Theory class on agricultural pollution, water and nutrient management; practical class on diffuse and point pollution sources using R. Academic contact for the Lisbon courses: Prof. Cláudia Marques-dos-Santos.

MSc supervision

Sofia Verde | Aarhus University

Co-supervision with Dr. Morten Graversgaard. Thesis: “Alternative protein transition of pulses in Denmark: exploring barriers, opportunities and pathways for pulses in the alternative proteins transition and a multi-level perspective on stakeholders’ perceptions”.

PhD mentoring

University of Lisbon

Joana Marinheiro (TERRA, ISA-UL, main supervisor Prof. Cláudia Marques-dos-Santos) in conducting emergy assessments for montado (*dehesa*) systems in Portugal.

Department of Agroecology and Land-CRAFT, Aarhus University.

Xiao Bai (Land-CRAFT, main supervisor Prof. Klaus Butterbach-Bahl) in upscaling GHG measurements (nitrous oxide, methane and carbon dioxide) in urban green spaces in Aarhus.

Tom Cripps (Land-CRAFT, main supervisor Prof. Klaus Butterbach-Bahl) in building a predictive framework using machine learning model.

Meshach Ojo Aderele (Land-CRAFT, main supervisors Prof. Klaus Butterbach-Bahl and Dr. Jaber Rahimi) in building a surrogate model of LandscapeDNDC to optimize Danish cropping systems.

Sevval Frank (Land-CRAFT, main supervisor Prof. Klaus Butterbach-Bahl) in landscape optimization of cropping systems and the development of an agricultural digital twin for Denmark's farm vocational schools.

School for International Training

Scientific committee member in Portugal, 2021/2022–2023/2024, focused on agriculture and the environment. Academic contact: Prof. Cátia Magro. <https://www.sit.edu/about-us/>

Scientific advising of student projects:

2024/2025

Nitrogen pollution in Portugal and Europe, with emphasis on groundwater nitrate contamination and Nitrate Vulnerable Zones.

2023/2024

Implementation of syntropic farming in New Mexico using lessons from Portuguese case studies; challenges and solutions for agricultural practices in Northern Portugal using performance indicators.

2022/2023

Environmental impacts of wine production in the Douro; carbon and water footprints in two wineries; effects of urban tree canopies on quality of life and thermal comfort.

Presentations

Invited presentations

1. **Serra, J.**, Quemada, M., Dobermann, A., Haefele, S., Poole, C., Lammell, J., 2026. EUNEP – Towards an EU crop-specific Safe Operating Space. Plant Nutrition Committee Fertilizers Europe, 1 September 2026 (Private).
2. **Serra, J.**, Quemada, M., Dobermann, A., Haefele, S., Poole, C., Lammell, J., 2026. EUNEP – Towards an EU crop-specific Safe Operating Space. FAO Cropland Nutrient Balance team, 4 August 2026 (Private).
3. **Serra, J.**, Quemada, M., Dobermann, A., Haefele, S., Poole, C., Lammell, J., 2026. The vision of EUNEP. GPNM-INI Webinar, 22 May 2026. Link: <https://www.youtube.com/watch?v=ix3T6QL65HE>

Conferences and workshops

1. **Serra, J.**, Marinheiro, J., Marques-dos-Santos, C., Lassaletta, L., Aguilera, E., Vries, W., Dalgaard, T., Hutchings, N. J., Einarsson, R., Graversgaard, M., Quemada, M., Giannini-Kurina, F. (2024). Paving the path towards a better understanding of N management in agriculture. XXII N Workshop 2024, Aarhus, Denmark.
2. **Serra, J.**, Cameira, R., Marques-dos-Santos, C., Lassaletta, L., Aguilera, E., Sanz-Cobena, A., Giannini-Kurina, F., Garnier, J., de Vries, W., Dalgaard, T., Hutchings, N. J., Einarsson, R., Graversgaard, M., Medinets, S., Quemada, M. (2024). Assessing groundwater nitrate hotspots in Europe reveals severely inadequate Nitrate Vulnerable Zones. XXII N Workshop 2024, Aarhus, Denmark.
3. **Serra, J.**, Cameira, R., Marinheiro, J., Marques-dos-Santos, C., Lassaletta, L., Garnier, J., Aguilera, E., Sanz-Cobena, A., de Vries, W., Billen, G. (2024). Nitrogen inputs by irrigation water sources is a missing link in the agricultural nitrogen cycle and related policies in Europe. 9th International Nitrogen Conference (N-2024), New Delhi, India.
4. **Serra, J.**, Cameira, R., Marques-dos-Santos, C., Lassaletta, L., Aguilera, E., Sanz-Cobena, A., Garnier, J., de Vries, W., Dalgaard, T., Hutchings, N. J., Einarsson, R., Graversgaard, M., Medinets, S., Quemada, M. (2024). Assessing groundwater nitrate hotspots in Europe reveals severely inadequate Nitrate Vulnerable Zones. 9th International Nitrogen Conference (N-2024), New Delhi, India.
5. **Serra, J.**, Cameira, R., Marques-dos-Santos, C., Lassaletta, L., Aguilera, E., Sanz-Cobena, A., Garnier, J., de Vries, W., Dalgaard, T., Hutchings, N. J., Einarsson, R., Graversgaard, M., Medinets, S., Quemada, M. (2023). Assessing groundwater nitrate hotspots in Europe reveals severely inadequate Nitrate Vulnerable Zones. RAMIRAN 2023, 18th International Conference, Cambridge, UK. Book of abstracts can be found in <https://resources.rskgroup.com/view/535287122/>

6. **Serra, J.**, Cameira, R., Marinheiro, J., Marques-dos-Santos, C., Lassaletta, L., Garnier, J., Aguilera, E., Sanz-Cobena, A., de Vries, W., Billen, G. (2022). Nitrogen inputs by irrigation water sources is a missing link in the agricultural nitrogen cycle and related policies in Europe. XXI Nitrogen Workshop 2022, 23-27 October 2022. Madrid, Spain.
7. **Serra, J.**, Cameira, R., Marinheiro, J., Marques-dos-Santos, C., Lassaletta, L., Garnier, J., Aguilera, E., Sanz-Cobena, A., de Vries, W., Billen, G. (2021). European-wide nitrogen input from irrigation water sources: Implications and outlook for nutrient management. AGU Fall Meeting Abstracts 2021, 13-17 December, 2021. New Orleans, United States of America
8. **Serra, J.**, Marques-dos-Santos, C., (2019). Overview of environmental N losses from agriculture in Portugal, with an emphasis on groundwater pollution. Invited speaker for the 19th Join Expert Group Meeting on Dynamic Modelling (JEG DM), October 30 – November 1, 2019, Sitges, Spain
9. **Serra, J.**, Marques-dos-Santos, C., Cruz, S., Hutchings, N. J. (2019). Importance of the contribution of nitrogen in irrigation water to gross nitrogen balances and pollution hotspots in a Mediterranean context. Ecology across Borders, Embedding Ecology in Sustainable Development Goals, 29 July – 2 August 2019, Lisbon, Portugal.
10. Cruz, S., Carnell, E., Pinho, P., Marques-dos-Santos, C., **Serra, J.**, Dragosits, U., Skiba, U. (2019). How data spatial resolution influences the usefulness of statistical data: a case study of N₂O emissions from agriculture in a Mediterranean country. Red REMEDIA Workshop 2019, 27-28 March, Lugo, Spain.
11. **Serra, J.**, Marques-dos-Santos, C., Cruz, S., Hutchings, N. J. (2019). Overview of environmental N losses in Portugal: a NitroPortugal synthesis. Red REMEDIA Workshop 2019, 27-28 March, Lugo, Spain.
12. **Serra, J.**, Marques-dos-Santos, C., Cruz, S., Hutchings, N. J. (2018). Effectiveness of disaggregated N balances at different spatiotemporal scales. NitroPortugal Wrap up Presentation, 17 November 2018, Lisbon, Portugal
13. Marques-dos-Santos, C., Cruz, S., Rego, C., Dalgaard, T., Sutton, M., **Serra, J.** (2018). After NitroPortugal – The Implementation of Nitrogen Footprint concept at the farm level in NEP. International Congress on Grapevine and Wine Sciences (ICGWS), 7-9 November 2018, Logroño, Spain.